

## Questions (47)

### 1 - Quiz

What is the main difference between `np.arange(0, 1, 0.1)` and `np.linspace(0, 1, 11)`?

▲ `arange` always includes the endpoint, `linspace` does not

● `arange` is slower

◆ `linspace` generates evenly spaced values including both endpoints (0 and 1) ✓

■ `linspace` only works with integers

### 2 - Quiz

When do we use the operator `@` between NumPy arrays?

▲ For simple concatenation

● Only for dot product between two vectors

◆ For elementwise multiplication

■ For matrix multiplication following linear algebra rules ✓

### 3 - Quiz

Why can the expression `a == b` be incorrect for floating-point comparison?

▲ Values are stored with finite precision, leading to rounding difference ✓

● There is a floating-point bug in Python

◆ Python compares memory addresses

■ Floats are silently converted to strings

### 4 - Slide

Correct recommended alternative

`np.isclose(a, b)`

### 5 - Quiz

What is the main advantage of Horner's rule for evaluating a polynomial?

▲ It gives an approximate result

● It computes  $p(x)$  without multiplications

◆ It reduces computation from  $O(n^2)$  to  $O(n)$  operations ✓

■ It avoids needing coefficients

### 6 - Quiz

What property characterizes the Lagrange basis functions  $L_i(x)$ ?

▲  $L_i(x_j) = x_i \cdot x_j$

●  $L_i(x_j)$  is always random

◆  $L_i(x_j) = 2$  for all  $i \neq j$

■  $L_i(x_j) = \delta_{i,j}$  ✓

### 7 - Quiz

Why is the barycentric formula preferred over the naïve Lagrange formulation?

▲ It changes the interpolation result

● It evaluates the polynomial in  $O(n)$  time and is numerically stable ✓

◆ It does not require weights

■ It computes derivatives automatically

### 8 - Quiz

Why does solving interpolation through the Vandermonde system become numerically unstable for large  $n$ ?

▲ Vandermonde matrix becomes highly ill-conditioned as nodes get close ✓

● The polynomial degree becomes too small

◆ Lagrange polynomials do not exist for large  $n$

■ Float values cannot be used inside matrices in python

### 9 - Quiz

What is the complexity of barycentric interpolation if you evaluate  $p(x)$  at **many** query points while nodes remain fixed?

▲  $O(n^2)$  preprocessing for weights and  $O(n)$  per evaluation ✓

◆  $O(n^3)$  preprocessing and  $O(\sqrt{n})$  per evaluation

●  $O(n^3)$  per evaluation

■  $O(n)$  preprocessing and  $O(1)$  per evaluation

### 10 - Quiz

Why does trigonometric interpolation avoid the “endpoint instability” seen in polynomial interpolation?

▲ Because trigonometric polynomials have smaller coefficients

◆ Because sampling is noiseless

● Because we ignore values near endpoints

■ Because periodic endpoints match ( $0 \equiv 1$ ), so no boundary tension occurs ✓

### 11 - Quiz

Given  $N$  equally spaced samples, what is the maximum number of distinct frequencies we can uniquely represent?

▲  $N$

◆  $N-1$

● About  $N/2$  ✓

■  $\sqrt{N}$

### 12 - Quiz

Why do frequencies  $k$  and  $N-k$  become indistinguishable after sampling at  $t_j = \frac{j}{N}$ ?

▲ Their cosine coincide, and sine samples differ only by sign (aliasing) ✓

◆ Both become zero for all samples

● Sin disappears numerically

■ Because the signal becomes constant

### 13 - Quiz

Why is FFT dramatically faster than direct DFT evaluation?

▲ It approximates instead of computing exactly

◆ It ignores high frequencies

● It reduces complexity from  $O(N^2)$  to  $O(N \log N)$  ✓

■ It reduces the number of samples

### 14 - Quiz

What is the computational cost of the naive DFT implementation?

▲  $O(N)$

◆  $O(N \log N)$

●  $O(N^2)$  ✓

■  $O(\log N)$

### 15 - Quiz

What is the relationship between the FFT and the DFT?

▲ FFT computes the exact same result as the DFT ✓

◆ FFT computes a different transform

● FFT requires real signals only

■ FFT produces an approximation of DFT

### 16 - Quiz

What happens when you multiply DFT coefficients by a mask and compute the inverse transform?

▲ You randomize the signal

◆ You filter the signal in time domain ✓

● You permute its samples

■ You increase its frequency content

## 17 - Quiz

Why are the vectors  $(1, w_N^k, w_N^{2k}, \dots)$  ?

▲ They are orthonormal under the standard dot product

◆ They are eigenvectors of shift matrix and hence of all circulant matrices ✓

● They form eigenvectors of all diagonal matrices

■ They maximize DFT magnitude

## 18 - Quiz

The trapezoidal rule integrates exactly polynomials of which degree?

▲ Up to degree 1 ✓

◆ Only constants

● Only quadratics

■ All polynomials

## 19 - Quiz

Quadrature: If halving the step size divides the error by about 4, what is the order of the rule?

▲ 1

◆ 2 ✓

● 4

■ not enough info

## 20 - Quiz

Quadrature: What happens to global error when local error is  $O(h^{p+1})$  ?

▲ It becomes zero

◆ It becomes  $O(h^p)$  ✓

● It becomes  $O(h^{p+2})$

■ It becomes constant

## 21 - Quiz

What is the main reason iterative methods are used for nonlinear equations?

▲ They give exact algebraic formulas

◆ They always converge to all roots

● We can solve them in one step

■ Most nonlinear problems do not have closed-form ✓

## 22 - Quiz

Which method guarantees convergence if the function is continuous and changes sign over an interval?

▲ Newton's method

◆ Bisection Method ✓

● Secant Method

■ Fixed-point iteration

## 23 - Quiz

Which condition guarantees local convergence of fixed-point iteration?

▲  $\Phi(x)$  is strictly increasing

◆  $|\Phi'(x^*)| < 1$  ✓

●  $|\Phi(x)| < 1$

■  $f(x)$  is monotone

## 24 - Quiz

If Newton's method converges locally, how fast is it?

▲ Linear

◆ Exponential ✓

● Sublinear

■ Quadratic

## 25 - Quiz

Why does Newton's iteration fail when  $f'(x_k) \approx 0$

▲ The update term explodes ✓

◆ The convergence becomes too fast

● The method becomes linear

■ The fixed point disappears

## 27 - Quiz

What is the asymptotic convergence order of the secant method?

▲ 1

◆ 1.618 ✓

● 2

■ 3

## 29 - Quiz

Why does Newton's method become **quadratically convergent**?

▲ Because it stores past iterates

◆ It forces symmetry of residual ✓

● Because its step is very small

■ Its iteration  $\Phi(x^*) = 0$

## 31 - Quiz

Why do we avoid computing the inverse matrix explicitly when solving linear systems?

▲ Computing the inverse is impossible

◆ Inverses only exist for diagonal matrices

● Python cannot invert matrices

■ Solving with LU factorization is faster and numerically safer ✓

## 26 - Quiz

Why is the secant method considered a derivative-free version of Newton?

▲ It ignores function evaluations

◆ It uses higher-order expansions

● It estimates  $f'$  using two previous iterates ✓

■ It is used only when  $f$  has no derivative

## 28 - Quiz

Which property guarantees that the **bisection method** will find a root inside an interval  $[a,b]$ ?

▲ The function is quadratic

◆ The interval is small

●  $f(a) \cdot f(b) < 0$  ✓

■  $f'(x) \neq 0$  everywhere in  $[a,b]$

## 30 - Quiz

What is the main practical advantage of Quasi-Newton/Broyden over full Newton?

▲ It converges faster

◆ It never needs a linear solver

● It avoids computing full Jacobians every step ✓

■ It converges even when no root exists

## 32 - Quiz

What is the computational complexity of Gaussian Elimination?

▲  $O(n^3)$  ✓

◆  $O(n)$

●  $O(n^2)$

■  $O(n^2 \log n)$

## 33 - Quiz

Computing  $PA=LU$  and solving for multiple right-hand sides  $b_1, b_2, \dots, b_l$  (for  $l \ll n$ ) costs

▲  $O(l \cdot n^3)$

◆  $O(n^3)$  ✓

●  $O(l \cdot n)$

■  $O(l^2 \cdot n^2)$

## 35 - Quiz

The Sherman–Morrison–Woodbury formula is used when:

▲ The system becomes triangular

◆ A matrix has a low-rank modification ✓

● A matrix is diagonalizable

■ A system is symmetric

## 37 - Quiz

Why is `np.linalg.lstsq` preferred in Python?

▲ It uses SVD internally and is numerically stable ✓

◆ It computes the inverse directly

● It requires matrix  $A$  to be square

■ It only uses safe integer arithmetic

## 39 - Quiz

What is the projection of a vector  $b$  onto the subspace spanned by columns of  $Q$

▲  $Q^T b$

◆  $R^{-1}b$

●  $QQ^T b$  ✓

■  $Qb$

## 34 - Quiz

What is the Schur complement of  $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$

▲  $A_{11}A_{22}$

◆  $A_{21}A_{12}$

●  $A_{22} - A_{21}A_{11}^{-1}A_{12}$  ✓

■  $A_{12} - A_{11}A_{22}^{-1}A_{21}$

## 36 - Quiz

When does the formula  $x = (A^T A)^{-1} A^T b$  produce a unique least squares solution?

▲ Always

◆ Only when  $A$  is square

● When  $A$  has full column rank ✓

■ When  $b = 0$

## 38 - Quiz

Which operation often destroys sparsity most severely?

▲ scaling rows of  $A$

◆ multiplying by diagonal matrix

● adding identity

■ forming  $A^T A$  ✓

## 40 - Quiz

Why is Classical Gram–Schmidt numerically unstable?

▲ It uses too many multiplications

◆ It gives complex-valued vectors

● It only works for symmetric matrices

■ It can't subtract nearly parallel vectors ✓

## 41 - Quiz

If  $H(v) = I - 2 \cdot \frac{vv^T}{v^T v}$  then...

▲  $H(v)^{-1} = 2H(v)$

◆  $H(v)^T H(v) = I$  ✓

●  $\det(H(v)) = 0$

■  $\det(H(v)) = 1$

## 42 - Quiz

Given rotations are especially good when

▲ Matrix is diagonal

◆ Values are integers

● Matrix is dense

■ Matrix is sparse ✓

## 43 - Quiz

The singular values of A are equal to:

▲ Eigenvalues of A

◆ Diagonal entries of L

● Entries of R in QR

■ Square roots of eigenvalues of  $A^T A$  ✓

## 44 - Quiz

What do the last columns of V (those past the rank) span in the SVD decomposition

▲ Column space

◆ Nullspace of A ✓

● Left nullspace

■ Row space of A

## 45 - Quiz

Which vector maximizes  $\|Ax\|$  subject to  $\|x\|=1$

▲ The first right singular vector ✓

◆ Any eigenvector of A

● The last left singular vector

■ The first column of U

## 46 - Quiz

According to Eckart-Young, the error of the best rank-k approximation  $A_k$  satisfies

▲  $\|A - A_k\|_2 = \sum_{i=k}^{\min\{n,m\}} \sigma_i$

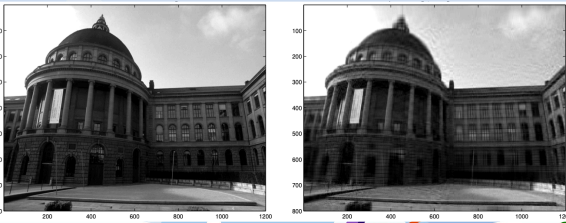
◆  $\|A - A_k\|_2 = \sigma_{k+1}$  ✓

●  $\|A - A_k\|_2 = \sigma_1$

■  $\|A - A_k\|_2 = \sigma_k^2$

## 47 - Quiz

What is the rank of the compressed photo?



▲ 40 ✓

◆ 300

● 5

■ 100

## Details

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